Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. An organization wants to ensure that data sent over the network cannot be read by unauthorized interceptors. Which security goal is primarily being addressed? (1 Marks)

A. Integrity
B. Availability
C. Confidentiality
D. Non-repudiation

Q2. Which of the following cryptographic approaches requires both parties to share a single secret key for both encryption and decryption? (1 Marks)

A. Asymmetric encryption
B. Symmetric encryption
C. Hashing algorithms
D. Public key infrastructure

Q3. An attacker sends an email that appears to be from a legitimate bank, directing users to a spoofed website to harvest credentials. What type of attack is this? (1 Marks)

A. Phishing
B. Man-in-the-Middle
C. Denial of Service
D. SQL Injection

Q4. Which access control model restricts resource access based on the sensitivity label of the resource and the clearance level of the user? (1 Marks)

A. Role-Based Access Control (RBAC)
B. Discretionary Access Control (DAC)
C. Mandatory Access Control (MAC)
D. Rule-Based Access Control

Q5. What is the primary purpose of adding a unique 'salt' value to passwords before hashing them? (1 Marks)

A. To encrypt the password so it can be decrypted later.
B. To prevent precomputed rainbow table attacks.
C. To compress the password size for faster database lookups.
D. To enforce complexity requirements on the user.

Q6. A security administrator notices unauthorized database queries originating from a web application input field. Which vulnerability is likely being exploited? (1 Marks)

A. Cross-Site Scripting (XSS)
B. SQL Injection (SQLi)
C. Cross-Site Request Forgery (CSRF)
D. Buffer Overflow

Q7. Which security concept dictates that users should only be granted the minimum level of access necessary to perform their job functions? (1 Marks)

A. Defense in Depth
B. Separation of Duties
C. Principle of Least Privilege
D. Need to Know

Q8. What type of malware encrypts a victim's files and demands payment in exchange for the decryption key? (1 Marks)

A. Spyware
B. Ransomware
C. Rootkit
D. Trojan Horse

Q9. In Multi-Factor Authentication (MFA), which of the following combinations represents two distinct authentication factors? (1 Marks)

A. A password and a PIN
B. A fingerprint scan and a retina scan
C. A smart card and a hardware token
D. A password and a one-time passcode sent to a registered phone

Q10. Which device inspects network traffic and actively blocks malicious packets based on predefined rules or signatures? (1 Marks)

A. Intrusion Detection System (IDS)
B. Intrusion Prevention System (IPS)
C. Network Hub
D. Proxy Server

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: C
Rationale: Confidentiality ensures that sensitive information is not disclosed to unauthorized individuals.

Q2. Correct Answer: B
Rationale: Symmetric encryption uses the same shared key for both encrypting and decrypting data.

Q3. Correct Answer: A
Rationale: Phishing is a social engineering attack designed to steal user data such as login credentials.

Q4. Correct Answer: C
Rationale: Mandatory Access Control uses security labels and clearances to enforce access policies.

Q5. Correct Answer: B
Rationale: Salting ensures identical passwords produce different hashes, neutralizing rainbow table attacks.

Q6. Correct Answer: B
Rationale: SQL Injection occurs when untrusted user input is directly concatenated into database queries.

Q7. Correct Answer: C
Rationale: The Principle of Least Privilege limits user privileges to only what is essential for their work.

Q8. Correct Answer: B
Rationale: Ransomware holds digital assets hostage by encrypting them until a ransom is paid.

Q9. Correct Answer: D
Rationale: This combination uses 'something you know' (password) and 'something you have' (phone).

Q10. Correct Answer: B
Rationale: An IPS actively prevents attacks by blocking malicious traffic in real-time.